A Three Dimensional DRAM (3D DRAM) Technology for the Next Decades

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Abstract

Three dimensional structured DRAM technology has drawn huge attention recently for its potential to fulfill high speed operation and low power consumption. In this paper, 3D DRAM with vertical bit line (BL) architecture is introduced as a promising solution to overcome scaling limitation for future DRAM technology. Full chip integration with 5-layered cell stacked on peri-core wafer is successfully demonstrated for the first time, offering superior on-current performance and gate controllability. A novel process integration scheme using Si/SiGe sacrificial multilayers and hybrid wafer bonding technique is presented with excellent full chip operation of 3D DRAM.

Introduction

Conventional DRAM (2D DRAM) is currently facing significant barriers in scaling beyond sub 10nm node. Therefore, 3D DRAM structure with vertically stacked memory cells has been considered as a potential candidate for the next DRAM era where higher density and lower cost are required [1]. Based on vertical word line (WL) architecture, integration scheme, circuit simulation, and cell transistor characteristics have been reported [1] [2]. In this paper, we propose a novel 3D DRAM technology that includes multi-stacked cell integration with a vertical BL architecture and hybrid wafer bonding process to connect cell wafer to peri-core wafer. In addition, we demonstrates full chip 3D DRAM operation with characteristics, such as subthreshold slope, GIDL and retention time, that are compatible with those of conventional 2D DRAM.

Architecture and Technology

Fig. 1(a) shows equivalent circuit diagram of 3D DRAM with vertical BL architecture. Memory cell layers are connected vertically via BL, and laterally through WL. Isolation areas define unit cell separating each from neighboring cells and WL runs laterally through the space between isolation areas as indicated in Fig. 1(b).

Si/SiGe multilayer stacked by epitaxial growth constructs the backbone for high quality channel layers. Patterned isolation areas serve as supporting pillars. BL and capacitor holes are generated through Si/SiGe stack to selectively etch out sacrificial SiGe layers to define Si channel layers with 7nm thickness. Cell transistor dimension is defined by lateral trimming process. Gate dielectric deposition is followed by replaced metal gate process to complete lateral WL formation. After junction engineering to optimize cell transistor, silicide and vertical BL fill processes are completed subsequently. Lateral recess of Si layers through vertical capacitor holes fabricates cell capacitor in lateral direction. It is crucial to control Si recess to define capacitor area precisely in 3D DRAM structure.

By hybrid wafer bonding technology, stacked cell wafer is connected to peri-core wafer fabricated separately, delivering cell-on-peri (CoP) architecture. Back-end processes including backside CMP, self-insulated contact, and power line integration are processed in the conventional order.

Results and Discussions

Fig. 2 shows cross-sectional TEM image of 3D DRAM structure with vertical BL. Stacked cell arrays on peri-core wafer are successfully integrated using hybrid wafer bonding technology. As shown in Fig. 3, resistance of chain formed across the interface between cell and peri-core wafers confirms high quality of hybrid bonding process. To form WL interconnection, as shown in Fig. 4, stair-like contact structure is fabricated on 5 cell layers by adapting integration technique developed in 3D NAND technology [3]. To accommodate more cell stacks for higher bit density, elaborate design of WL connection and relevant process need to be developed.

Fig. 5 shows capacitance modulation of integrated 3D DRAM with various process options such as structural dimension of capacitor and optimization of dielectric materials. Performance of cell transistors is shown in Fig. 6. Double gate cell structure demonstrates excellent subthreshold slope of 63mV/decade, effectively controlling channel in <10nm thickness range. Junction engineering and contact process are optimized to achieve sufficient on-current while suppressing GIDL as low as that of conventional 2D DRAM.

Fig. 7 shows data retention characteristics of the 3D DRAM compared with 2D DRAM. Inherent structural susceptibility from floating body effect (FBE) in 3D DRAM cell transistors degrades data retention time in dynamic mode where more charge loss occurs through the floating body especially at low BL potential. Optimized GIDL by junction engineering in this work could effectively mitigate FBE and ensure retention time in dynamic mode. The broad distribution indicates that process uniformity among cells needs to be improved to enhance retention characteristics.

Conclusions

As a candidate to provide higher array efficiency and bit density in sub 10nm DRAM technology, a novel 3D DRAM with vertical BL architecture is introduced with key fabrication techniques. 5 cell layers were successfully integrated on peri-core wafers, demonstrating full chip operation of 3D DRAM for the first time. FBE in 3D DRAM was effectively suppressed by junction optimization, yielding retention characteristics comparable to 2D DRAM.

References

[1] J.W. Han, et al., Symposium on VLSI Tech. Dig., TFS1-1, 2023
[2] M. Huang, et al., International Memory Workshop, pp. 1-4, 2023.
[3] H. Tanaka, et al, VLSI Symp. Tech. Dig., pp14-15, 2007.

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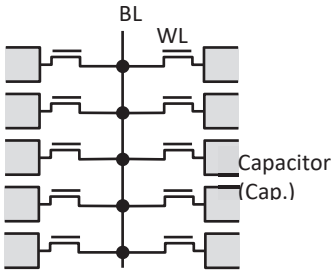
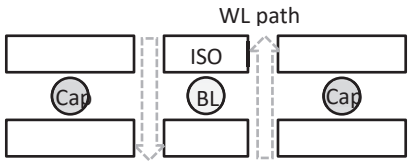


(a) Equivalent circuit of 3D DRAM cell layers



(b) Unit cell layout

Fig.1 (a) Equivalent circuit diagram and (b) Cell layout of 3D DRAM

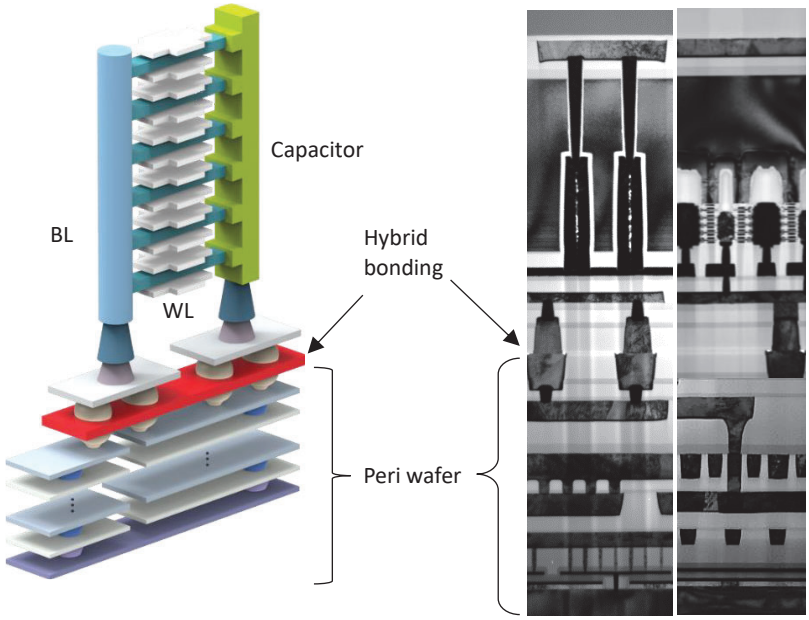


Fig.2 (a) Conceptual illustration and cross-sectional TEM of 3D DRAM with vertical BL architecture

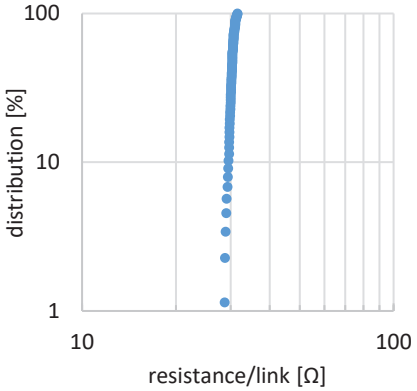


Fig.3 Chain resistance distribution of interface formed by hybrid bonding

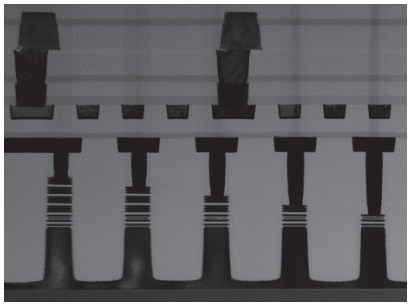


Fig.4 TEM image of WL contact area

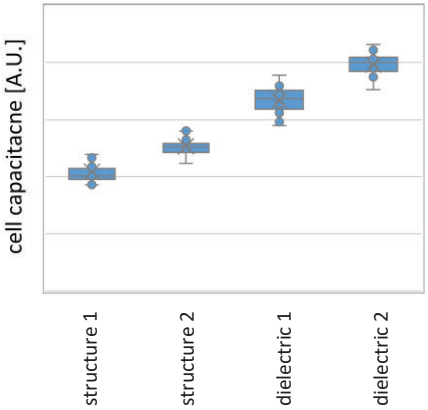
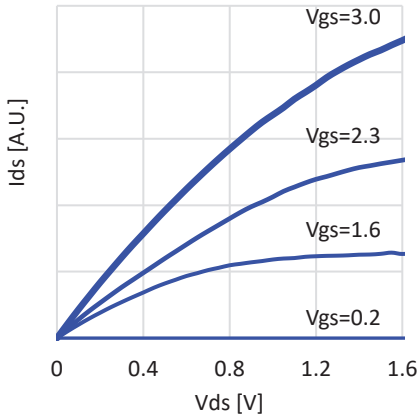
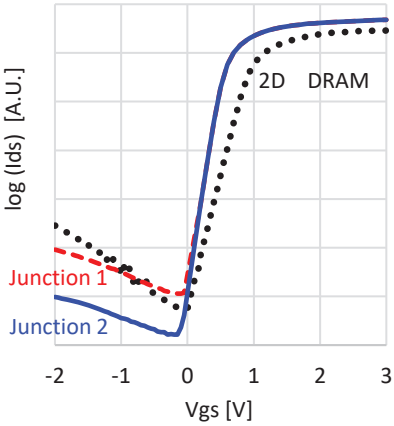


Fig.5 Cell capacitance with various process options



(a)



(b)

Fig.6 I-V characteristics of cell transistor (a) Id-Vd (b) Id-Vg

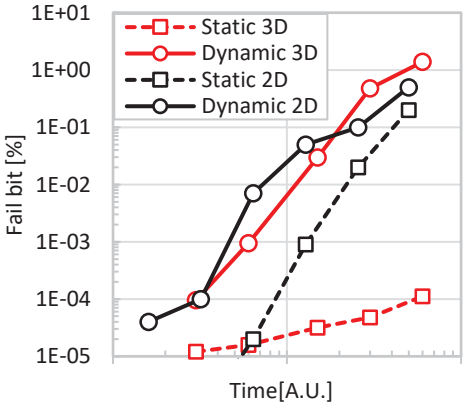


Fig.7 Data retention characteristics.

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